Lab-2: 8x1 Multiplexer and 8x3 Encoder

Aim:

To design and simulate an 8x1 Multiplexer and an 8x3 Encoder circuit using logic gates in Logisim and verify their operation.

Theory:

8x1 Multiplexer (MUX) is a combinational circuit that selects one of 8 input lines (I0–I7) and forwards it to a single output, based on 3 select lines (S0, S1, S2).

- Total inputs: 8 data inputs + 3 select lines → 1 output
- Boolean Expression: $Y = \bar{S}2 \cdot \bar{S}1 \cdot \bar{S}0 \cdot I0 + \bar{S}2 \cdot \bar{S}1 \cdot S0 \cdot I1 + \ldots + S2 \cdot S1 \cdot S0 \cdot I7$

8x3 Encoder is the reverse operation — it takes 8 input lines (one active at a time) and encodes the active line into a 3-bit binary output (A2, A1, A0).

- Total inputs: 8 lines → 3-bit binary output
- Boolean Expressions:
  - A0 = I1 + I3 + I5 + I7
  - A1 = I2 + I3 + I6 + I7
  - A2 = I4 + I5 + I6 + I7

Procedure:

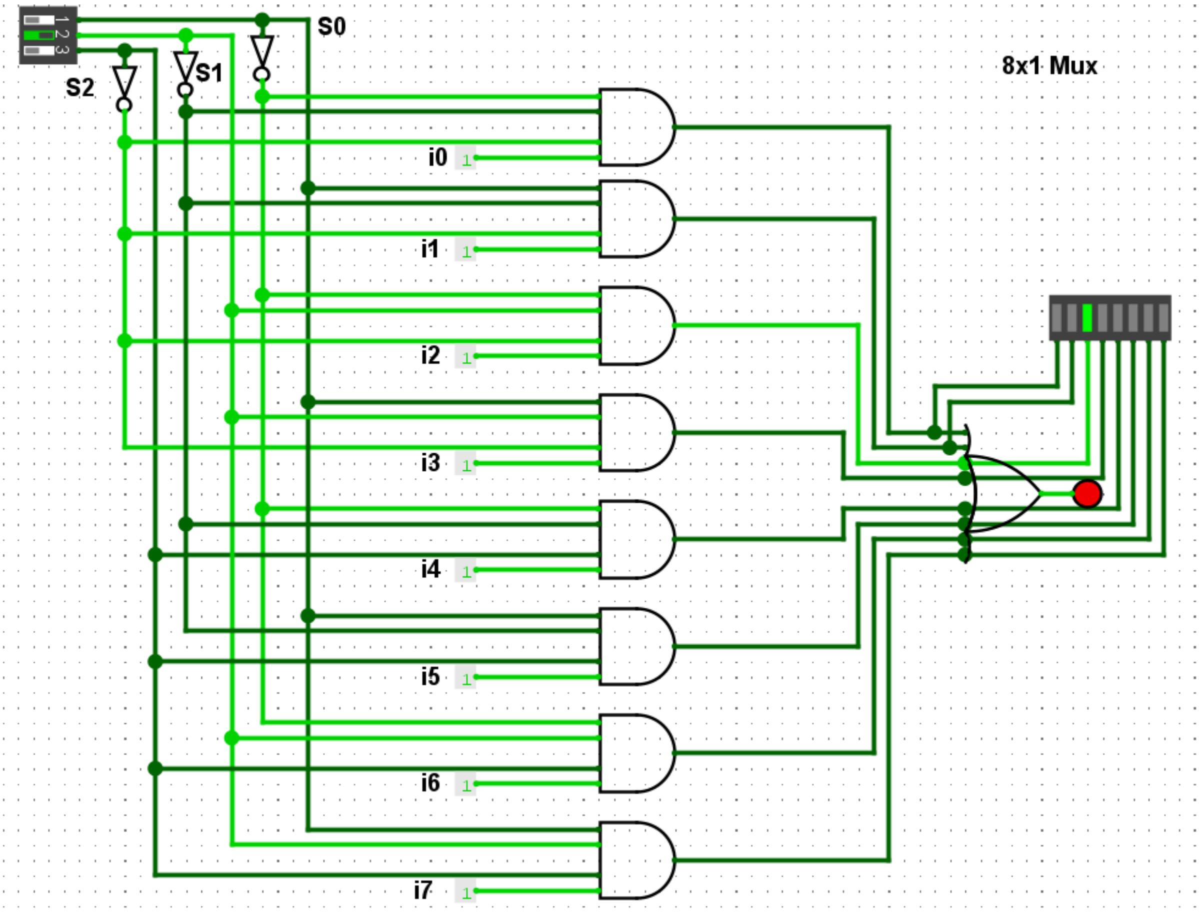
1. Open Logisim and create a new project.
2. 8x1 MUX: Place 8 AND gates (each with 4 inputs), connect data inputs I0–I7 and select lines S0, S1, S2 with appropriate complements using NOT gates. Connect all AND gate outputs to an OR gate to get the final output Y.
3. 8x3 Encoder: Connect input lines to three OR gates as per the Boolean expressions for A0, A1, and A2.
4. Apply input combinations and observe outputs for each circuit.
5. Verify outputs against the expected truth tables.

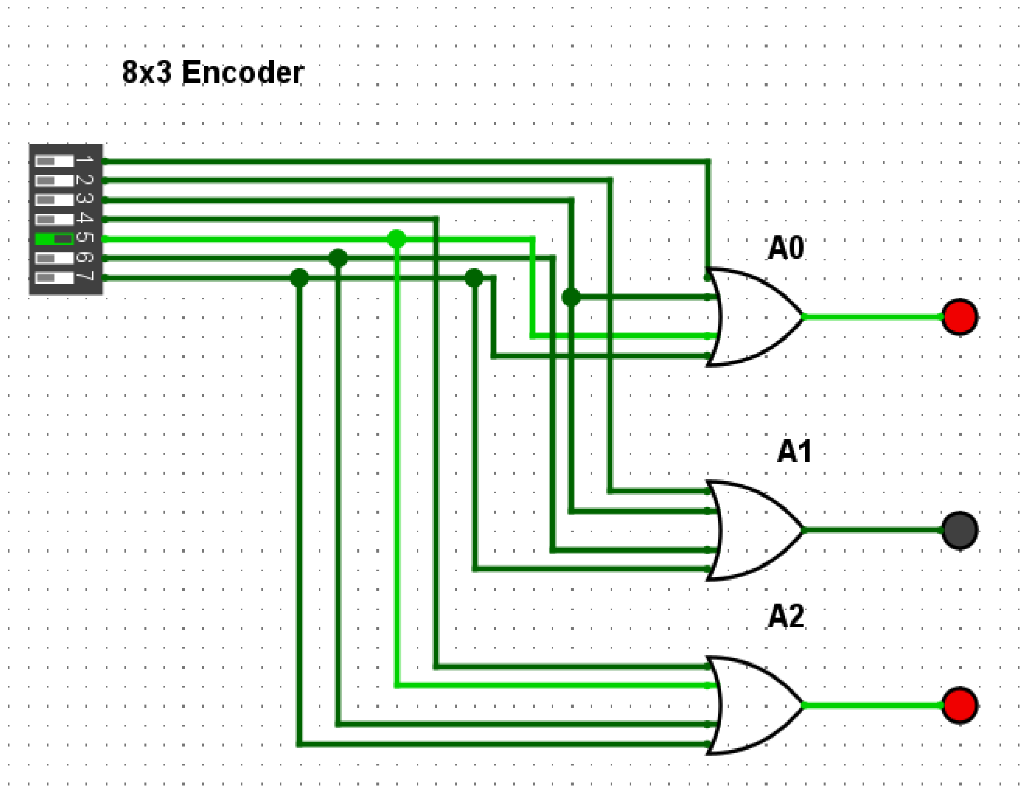
Table 1 8:1 Multipleexer Truth Table

Select Inputs			Outputs
S2	**S1**	**S0**	**Y**
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

Input									Output	
I7	I6	I5	I4	I3	I2	I1	I0	O2	O1	O0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Circuit Diagrams:





Conclusion:

The 8x1 Multiplexer and 8x3 Encoder circuits were successfully designed and simulated in Logisim. The MUX correctly routed the selected input line to the output based on the 3-bit select combination. The Encoder correctly produced a 3-bit binary code corresponding to the active input line. This experiment demonstrates the use of combinational logic for data selection and binary encoding.